

DIAGNOSIS BUYS A SHOPPING LIST, NOT A RATIO: BENCHMARK-GUIDED DATA REPAIR AND ITS GENRE- GATED RETURNS

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ABSTRACT

[TODO: abstract written last per C-12; thesis sentence source = CLAIMS CL-5: (i) genre determines the visibility of repair-data gains (+10–16pp in the repair genre vs +2–7pp on plain items); (ii) profile-matched ratios show no advantage in either genre — the actionable output of diagnosis is a shopping list (which components must be present, which are toxic, in which genre to validate), not a mixing ratio.]

1 INTRODUCTION

[TODO: written last (C-12). Spine: Fig. 1 invoked in paragraph 1; thesis = CL-5 two sentences; named alternative rejected = “diagnosis buys a mixing ratio”.]

2 DIAGNOSTIC AND REPAIR PROTOCOL

Our pipeline has three stages: a perturbation-probe suite that turns a benchmark score into a failure profile (§2.1), a library of repair-data components and mixing arms trained from that profile (§2.2), and a two-genre evaluation protocol with a fixed operating-point rule that separates repair gains from collateral damage (§2.3). Throughout, the subject is a single pre-repair model (Qwen3-8B instruct); all decoding is greedy with thinking mode disabled.

2.1 PROBE SUITE: FROM ONE SCORE TO A FAILURE PROFILE

For each GSM8K test item we generate a family of eight probes that hold the underlying arithmetic task fixed and perturb one interface at a time (Table 1). The original item (**O**) anchors the family. Three probes supply *assistance*: a step scaffold in verb-phrase form (**S**), an entity-free statement of the governing rule (**R**), and a correct intermediate value (**C**). Two probes plant *misinformation* in the context: a wrong intermediate value (**W1**) and a wrong final claim (**W2**); both record the planted value so that answering with it (*adopt*) can be distinguished from answering with any other wrong value (*derail*) or not answering (*mute*). One probe rephrases the surface text (**P**), and one requests the identical computation under a JSON output constraint (**F**). Assistance and misinformation probes are validated by leakage filters (scaffolds and rules must contain no digits and no problem entities); every probe type passed a manual construct audit of 100 items per type with a preregistered $\geq 70\%$ acceptance gate. The pool is GSM8K test (1,300 items) plus a filtered GSM-hard bucket; 330 items with negative or non-integer golds are excluded from probe construction.

A frozen classifier maps each item’s eight-bit signature to a failure bucket: *conduct* (solves O but is flipped by W1/W2), *format* (only F differs), *phrasing* (O and P disagree), *scaffold / rule* (fails O, recovered by S / by R alone), *local execution* (recovered only by C), and *unresolved* (fails everything). Mixed signatures go to an explicit mixed bucket rather than being forced into a single label. The resulting profile is the paper’s diagnostic object (Figure 1).

Probe	Perturbs	Holds fixed	Failure it isolates
O	— (original)	everything	baseline correctness
P	surface wording	task, numbers	phrasing sensitivity
S	+ step scaffold	task	missing-plan (scaffold) failures
R	+ explicit rule	task	missing-rule failures
C	+ correct intermediate	task	local execution failures
W1	+ wrong intermediate	task	credulity (conduct)
W2	+ wrong final claim	task	credulity (conduct)
F	output format (JSON)	task, wording	format-coupled computation

Table 1: The probe suite. Each GSM8K item yields one probe family; a frozen rule set maps each family’s pass/fail signature to a failure bucket.

2.2 REPAIR COMPONENTS AND MIXING ARMS

Each failure bucket names a candidate repair-data *component*; the components are deliberately minimal so that mixtures can be credited to their parts. **Conduct**: worked examples that verify a planted intermediate, reject it on mismatch, and recompute. **Format**: the same computation performed correctly *under* the JSON constraint. **Phrasing, scaffold, rule**: LLM-generated pools (kept 2,200 / 1,606 / 1,125 after validator gates with preregistered minimum-yield thresholds). **Drills**: bare “compute 17×23 ” items, included as a preregistered falsification component. **Replay**: clean GSM traces.

Arms are composed from these pools under a preregistered design (single-component arms: 600 items each, fixed; mixture arms: 2,000 items, character-aligned within 2%): **A1** mixes components in proportion to the measured failure prevalences (40/36/16/5/2 for conduct/format/phrasing/scaffold/rule); **B** mixes them uniformly; **D** applies the frozen reverse permutation of A1 (2/5/16/36/40); **C** is generic chain-of-thought data with no repair design; and **clean replay** (600 items) serves as the *training placebo*, isolating the effect of any fine-tuning at all. Dose arms replace N items of the fixed replay carrier with one component (format at 10/20/36%; drills at 10/25%), so the dose axis varies only the component share. All training runs use identical hyperparameters (full fine-tuning, $\text{lr } 10^{-5}$ cosine, epochs $\in \{2, 4, 8, 16\}$; A1, B and C are run with three seeds).

2.3 TWO-GENRE EVALUATION AND THE OPERATING-POINT RULE

Every checkpoint is evaluated in two genres. The plain genre re-runs the full probe suite: rescue is counted per bucket, in two frozen readings — the *strict* reading requires an item’s whole signature to turn clean; the *loose* reading requires only the bucket’s defining probes to flip from fail to pass — and every rescue number is reported as a delta against the clean-replay placebo, because placebo rescue is large in some buckets (§4). The repair genre presents 480 corrupted-context items (planted wrong steps, wrong claims, clean controls, and unanswerable cases) under a JSON decision contract; scoring reuses the frozen strict-abstain judge of the accounting study, so numbers are comparable across chapters.

Checkpoints are compared only at their *operating point*: the smallest epoch satisfying three health gates on plain items — answer rate ≥ 0.95 , JSON bleed-through $\leq 5\%$, mute rate $\leq 12\%$. The gate is applied per arm; when no epoch passes, the arm is reported as having *no clean operating point* rather than being scored at its best epoch. This rule is what separates a repair gain from a checkpoint that has merely traded general usability for the target metric.

3 THE DIAGNOSTIC PROFILE: WHAT A BENCHMARK SCORE HIDES

Applying the probe suite to the pre-repair model turns one number into a failure profile (Figure 1). The headline is a gap: the model scores 93.5 on GSM8K, but only 60.6% of items are answered correctly *and* survive every interface perturbation. The largest failure class is not inability but *credulity*: 16.4% of items are solved correctly in isolation yet flipped by a planted wrong intermediate or wrong final claim. Within that class the failure is overwhelmingly *derailment* rather than *adoption* (227 vs

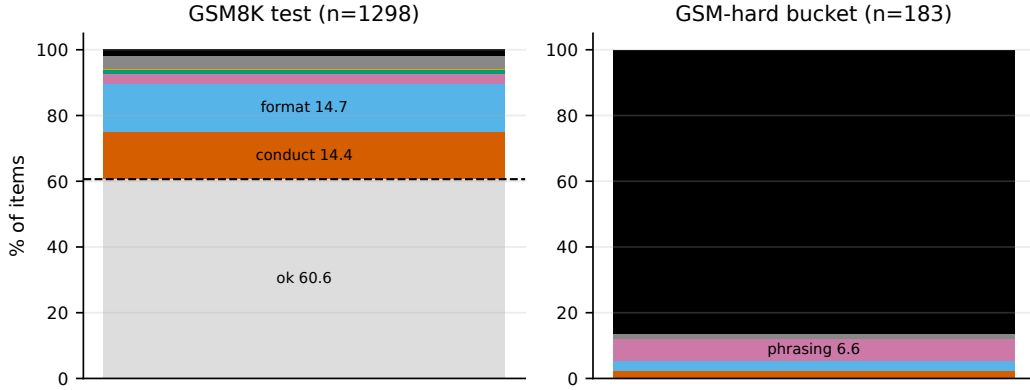


Figure 1: A benchmark score hides a failure profile. Left: of GSM8K’s 93.5 surface score, only 60.6% of items survive interface perturbation; the failure mass decomposes into credulity (conduct, 16.4%), format coupling (14.7%), phrasing sensitivity (6.4%), assistance-recoverable buckets (~3%), and unresolved (1.9%). Right: the GSM-hard bucket is dominated by unresolved failures (86.3%). Denominators, pool filters, and the labile-core discount are given in Appendix B.

9 items): the planted value knocks the computation off course, but the model almost never copies it verbatim. Format coupling is the second class (14.7%): the same computation requested under a JSON constraint fails, consistent with prior reports that format constraints degrade reasoning, here established by item-level controlled pairing.

Two properties of the profile matter for everything downstream. First, the buckets are not equally “real” as repair demand: a flip-overlap audit (Appendix, Figure ??) shows that 57% of the conduct bucket is rescued by *any* fine-tuning, including a clean-replay placebo — a deterministic labile core — while the format bucket has essentially no free core (1%). Second, the profile is pool-dependent: on GSM-hard, 86.3% of items fail every probe, so the taxonomy above describes the regime where the model is basically competent and fails at interfaces, not the regime where it cannot solve the task at all.

4 FROM PROFILE TO RECIPE: COMPONENTS MATTER, RATIOS DO NOT

The natural expectation — and our preregistered one — was that rescue tracks the match between component and bucket. It does not (Figure 2a). Measured against the placebo delta, only the format component shows a large specific own-bucket effect (+70–79pp in mixtures); conduct, phrasing, scaffold and rule sit within ± 16 pp of the placebo, whose own rescue floor is high (73% of the conduct bucket). Preregistered coefficients of .95 (conduct) and .90 (rule) were misses; the replacement picture is a high non-specific floor with sparse specific signal (Appendix A).

Mixture ratios follow the same logic (Figure 2b). The profile-matched mixture (A1), the uniform mixture (B), and the deliberately reversed mixture (D) are indistinguishable on plain-genre rescue, and in the repair genre matched ratios never exceeded uniform in any reading (B $.717 \pm .040$ vs A1 $.676 \pm .025$ over three seeds). What does matter is presence and toxicity: dropping the format component costs the entire format rescue, and the drills component — bare “compute X” items included as a falsification arm — is actively harmful, teaching the model to adopt planted values (W_{adopt} 25% at 100% dose) and sitting below the placebo on the conduct bucket.

Dose curves sharpen both statements (Figure 3). Format rescue saturates at a 10% dose — sixty items — and the toxicity of the pure diet does not creep in with dose: every mixed dose passes the health gates, and only the 100% diet is lethal. Drills harm is likewise not monotone in dose on plain items, but it is visible at a 25% dose in the repair genre (§5). The actionable output of diagnosis is therefore a *shopping list*, not a mixing ratio: add the specific components (cheaply), exclude the toxic ones, and let any clean data supply the floor.

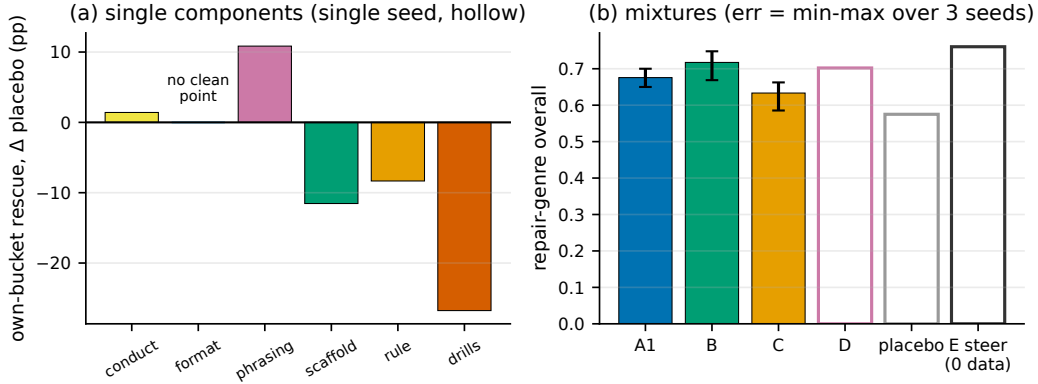


Figure 2: Components matter, ratios do not. (a) Own-bucket rescue of each single-component arm over the training placebo: only format shows a large specific effect; pure format has no clean operating point; drills is actively harmful. (b) Repair-genre accuracy of the mixture arms: repair-bearing arms separate from placebo and generic CoT, but profile-matched ratios (A1) never exceed uniform (B) in any reading. The zero-data steering anchor E (.760) exceeds every trained arm; its edge over placebo decomposes entirely into the two decision policies (keep +.41, abstain +.35). Error bars: min-max over 3 seeds; hollow bars: single seed.

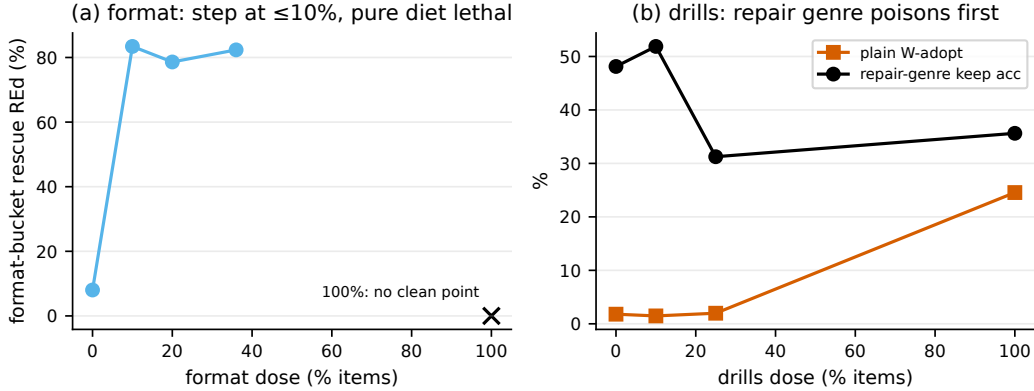


Figure 3: Dose curves. (a) Format rescue is a step, not a slope: a 10% dose (60 items) buys the full effect and only the 100% pure diet is lethal (mute-side collapse, no clean operating point). (b) Drills harm enters by genre: plain-genre adoption stays at carrier level through a 25% dose while repair-genre keep accuracy already collapses; at 100% both genres fail. Single seed.

5 GENRE GATING: THE (DOMAIN $\times$ GENRE) SWITCH

The same checkpoints tell different stories depending on the genre in which they are asked to perform, and the pattern recurs across three independent measurements (Figure 4). Repair-component gains that are barely visible on plain items (+2–7pp over placebo) separate cleanly in the repair genre (+10–16pp). Drills toxicity is invisible on plain items up to a 25% dose yet already collapses repair-genre keep accuracy by 17pp at that dose. The scaffold component rescues the format bucket on plain items while sitting 12pp *below* placebo in the repair genre.

The validation domain shows the gate is a genuine switch, not a one-way amplifier. On 2WikiMultihopQA (Figure ??a–b) the direction reverses: the anti-credulity component transfers on the plain face — adoption failures drop from 19 to 0 and derailments from 28 to 4 against a placebo at 7/56 — while repair-genre accuracy does not separate (+3.8pp, below the preregistered +8pp line, scored as a miss). Which genre reveals an effect is domain-dependent; what is constant is that no single genre suffices. The 2Wiki miniature also surfaced an unregistered cost specific to the knowledge domain:

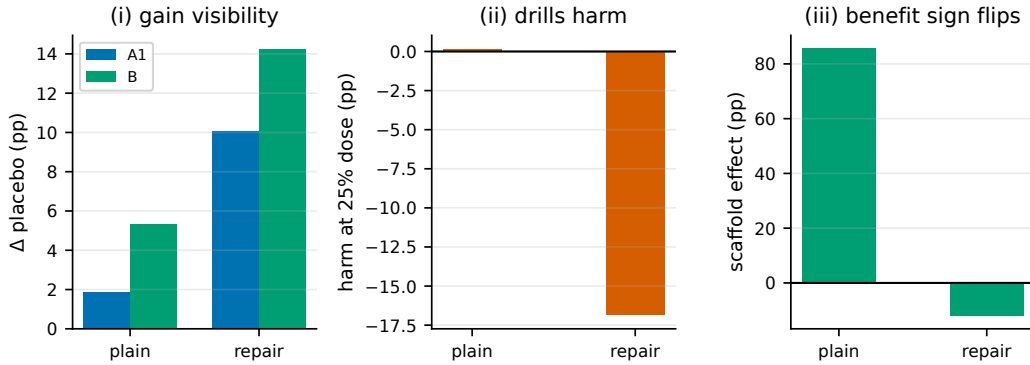


Figure 4: Genre gating: three independent measurements in which the evaluation genre switches the effect. (i) Repair-component gains are visible in the repair genre (+10–16pp over placebo) but not on plain items (+2–7pp). (ii) Drills harm at a 25% dose appears only in the repair genre. (iii) The scaffold component’s effect flips sign between genres (single seed; the structure-vs-JSON attribution is a known limitation).

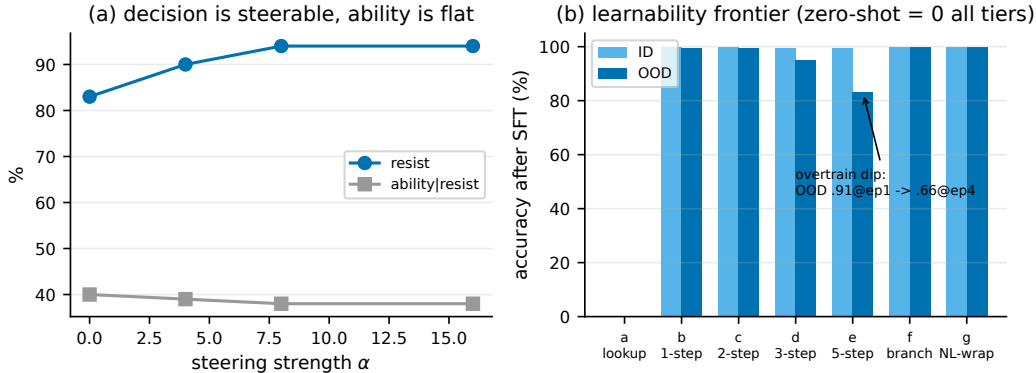


Figure 5: Mechanism and boundary. (a) A single activation direction moves the keep/update decision (resist 83→94%) while computation stays flat (ability|resist $\approx$ 38–40%). (b) The learnability frontier stays unbent: explicit invented rules are fully teachable and OOD-general through 3-step chains, 5-step chains (OOD 83%, with overtraining damaging OOD from its epoch-1 peak of 91%), conditional branching (100%), and natural-language wrapping (100%); zero-structure lookup is not (0.2%/0.0%).

every 600-item arm, including the placebo, damages known-item QA (5 → 34–56 new failures), and a transcript audit classifies 100% of those as answered-wrong with confabulated evidence chains — a content-side tax, distinct from the genre-bleed tax of the computation domain — and every arm lands below the pre-repair model on repair-genre accuracy. The prescription consequence is rule 2 of §7: validate in both genres, against the pre-repair model, without assuming a priori where the effect will show.

6 MECHANISM AND BOUNDARY

Two instruments locate the mechanism and the boundary. On the mechanism side, a single activation direction extracted from the pre-repair model’s own plain-genre probes moves the keep/update decision without moving computation (Figure 5a). The direction does not impose the repair genre on inputs that do not request it (plain-genre bleed 0), but it does raise schema compliance when the input explicitly requests one (JSON validity 30% → 75%): what it pushes is engagement with the requested repair task — decision plus task engagement — not indiscriminate genre injection. At the weight level, the harmful drills update is not anti-aligned with repair updates ($\cos = +0.13$, a

preregistered miss) but decorrelated: its alignment with every other component update is the global minimum (0.06–0.24 vs 0.4–0.6). “Reverse vaccine” is a behavioural description, not a geometric one.

On the boundary side, the learnability frontier refuses to bend (Figure 5b). Whatever makes GSM’s residual failures unrepairable by data, it is not chain depth (≤ 5 steps teachable), not control flow (branching 100%), and not linguistic wrapping (100%); the remaining suspects are task diversity and rule implicitness. The frontier experiments also yield a caution: overtraining damages already-acquired OOD generalisation (the 5-step tier peaks at epoch 1 and loses 25pp by epoch 4), the same early-landing/late-tax structure as the operating-point rule of §2.

The zero-data steering arm closes the loop (Figure 2b, E). Its repair-genre score (.760) exceeds every trained arm, and its entire edge over the placebo decomposes into the two decision policies (keep +.41, abstain +.35) with computation policies at placebo level. The reason the trained arms cannot match it is one we measured earlier at the data level: keep-style and override-style supervision are structurally mutually exclusive in a data mixture (the dose law), so SFT must choose and pay the keep collapse. The activation direction does not pay that tax. Interference that cannot be repaired at the data level can be bypassed at the activation level — the paper’s closing finding, and the reason steering is first-line for decision-class repairs in §7.

7 THE PRESCRIPTION MANUAL

Diagnosis pays for itself not by tuning mixture ratios but through five rules, each backed by a dose curve or a cross-genre reconciliation:

1. **Add the specific component; 60 items suffice.** Format rescue saturates at a 10% dose (a step, not a slope); further dose buys nothing.
2. **Validate effects and toxicity in both genres, against the pre-repair model; which genre reveals them cannot be assumed a priori.** On GSM the repair genre poisons first (drills at a 25% dose); on 2Wiki the plain face reveals the gain while repair-genre accuracy does not separate — the revealing genre is domain-dependent.
3. **The pure diet is the only lethal form.** 100% format data leaves no clean operating point (mute-side collapse); 100% drills teaches adoption of planted values (W_{adopt} 25%).
4. **Ratios do not matter; uniform coverage suffices.** Matched ratios never exceeded uniform in any reading.
5. **For decision-class repairs, steering is first-line; SFT is second-line.** Keep-style and override-style supervision are structurally mutually exclusive in a data mixture, so SFT pays a keep-collapse tax that an activation direction does not (E arm: .760, above every trained arm, entirely from the decision policies). SFT remains the choice when 99%+ reliability is required and the keep cost is acceptable; ability failures route to tools. The plain-genre adopt subset ($n=31$) is rescued by no data arm at this budget.

Every step of the manual states the genre in which it is validated (§5).

8 RELATED WORK

[TODO: differential lines incl. natural-set sources (one line each, neighbour-not-first discipline: GSM-IC↔W1, format-constraint literature↔CL-4).]

9 CONCLUSION

Diagnosis pays, but not in the currency we preregistered. A perturbation-probe profile of a strong model’s benchmark score buys a shopping list — which repair components must be present (format, sixty items), which are toxic (compliance-style drills), and in which genre, against which reference, each claim must be validated — rather than a mixing ratio, which never beat uniform in any reading. Effects are gated by (domain $\times$ genre); the teachability boundary excludes depth, control flow and

wrapping as the source of residual unrepairability; and the decision-level failure class that data mixtures repair only by paying a structural keep–override tax is repaired for free by a single activation direction. What data-level repair cannot disentangle, activation-level repair can bypass.

REFERENCES

A PREREGISTRATION SCORECARD

Every training experiment in this paper was preregistered before any run started; this table reports every prediction against its outcome, including the misses — the misses are where the paper’s replacement hypotheses come from.

Prereg	Prediction	Outcome	Verdict
Loop 3 (recipe)	conduct coefficient .95	specific effect +2pp over placebo	MISS
	rule coefficient .90	−9pp ($n=12$)	MISS
	format coefficient .60	no clean point (precondition failed)	MISS
	phrasing / scaffold .50±	51% / 42% raw	HIT / HIT (raw)
	drills ≈ 0	direction right, harm unpredicted	dir. HIT
	A2 arm (measured ratios)	inputs falsified in 4/5 buckets	non-executable
Batch 2 (dose)	all format doses pass gates; saturate by 20%	all pass; saturates at $\leq 10\%$	HIT*
	drills harm monotone (≥ 5 pp at 10%)	flat on plain items; genre-gated	MISS
	scafffimt separates structure vs JSON	falls in JSON interval (attribution stayed)	HIT (rule)
Loop 5 (ΔW)	$\cos(d, \Delta W_{\text{drills}}) < 0$ single-component ΔW angles	+0.13: decorrelated, not opposed drills = global outlier (all pairwise min)	MISS registered
Tier 2 (frontier)	lookup tier A ≈ 0 ; depth decay M calibration formula	0.2/0.0%; monotone but no bend ≤ 3 steps self-neutralised at ceiling (no range)	5/5 HIT instrument note
Bendpoint (G/B)	5-step teachable (OOD $\geq 85\%$)	G<B; OOD .83, overtrain dip from .91	HIT (edge)
	branching = bend candidate	100%/100% — frontier unbent	MISS
	NL wrapping costs ≥ 20 pp	100%/100% — wrapping is free	MISS
E arm (steering)	conduct bucket $\approx 94\%$ repair genre $< +10$ pp	68% < placebo (resist-vs-correct mismatch) +42.7pp, above all trained arms	MISS MISS*
E1 (data size)	targeted $N \leq 300$ suffices; random unstable	mute-wave artifact; measure replaced	hard stop

Table 2: Preregistration scorecard. *as with format saturation, the preregistration undershot in the direction of the cheaper method — the miss direction is itself information; the E-arm result passed a four-point hard-stop autopsy before entering any figure (Appendix B). The Loop-3 misses motivated the placebo-delta reading (COR-1); the dose miss became the genre-gating result (§5); the ΔW miss reframed “reverse vaccine” as behavioural-only (COR-3).

Abstain variance warning. The retrieve-or-abstain policy is the highest seed-variance column in the study (uniform arm across three seeds: .34/.15/.45); a single-seed ordering anomaly ($D > B > A1$) collapsed entirely into this band and is recorded as noise, not finding.

B INSTRUMENTS AND AUDITS

[TODO: abstain seed-variance warning table; flip-overlap matrix; construct audit protocol; judge definitions (strict/loose readings, non-dict guard).]